

### 3. LINEAR PROGRAMMING

**MBS**

#### NUMERICAL QUESTIONS

1. 2071 Q.No. 11

The advertisement department of a certain firm wishes to plan its advertising strategy to reach certain minimum percentages of high and low income groups. Two alternatives are considered. Radio and Television. Radio advertising has an exposure for high income group of 2% per programme but only 1% exposure for low income group. Television on other hand expose 3% of low income group per show and only 1% of high income group per show. Radio advertising costs Rs.1,000 per programme and television Rs.4,000 per show. If firm wants a minimal exposure of 50% of high income group and 30% of low income group, what strategy should it use for minimizing advertising cost? Formulate the problem and solve it by using simple method.

Ans: Min.  $Z = \text{Rs. } 30,000$  at advertisement on radio = 30 and on television = 0

2. 2070 Old Q.No. 7

Solve the following linear programming problem and interpret the result using simplex method.

Maximize profit  $Z = \text{Rs. } 4,000X + \text{Rs. } 3,200Y$

Subject to the constraints

$$2X + Y \leq 30$$

$$2X + 5Y \leq 50$$

$$2X + 3Y \leq 38$$

Where  $X, Y \geq 0$  non-negative conditions.

Ans: Max  $Z = \text{Rs. } 64,800$ ;  $X = 13$ ;  $Y = 14$

3. 2069 (Old) Q. No. 7

A company combines factors P and Q to form a product which must weigh 500 kgs. At least 200 kgs of P and no more than 400 kgs of Q can be used. The cost of P is Rs. 250 per kg and of Q is Rs. 100 per kg. Formulate the problem in a mathematical form and use simplex method to find the amount of factor P and Q, which should be used to minimize the cost.

Ans: Min.  $Z = \text{Rs. } 80,000$ ;  $P = 200 \text{ kg}$  and  $Q = 300 \text{ kg}$



4. 2068 Q.No. 11

A company produces some tables and chairs. The profit contribution on each table is Rs 64 and on each chair Rs 80. Each table requires in its construction 20 units of special type of wood while each chair requires 40 units of same type of wood in its construction. The supplier can furnish the company with a maximum of 600 units of wood each day. Only 200 man hours of wood working are available to the company each day. Each table requires 20 hours and each chair requires 8 hours of wood working labour. The final step in the production process the finishing work is performed by highly skilled employees and a maximum of 114 man hours of finishing labour is available each day. Each table requires 9 man hours and each chair requires 6 man hours of finishing time. Formulate the problem and solve the given problem by using simplex method in order to maximize the profit contribution. [20]

Ans: Total max profit = Rs.1296; No. of chairs ( $x_2$ ) = 13; No. of tables ( $x_1$ ) = 4

5. 2068 Old Q.No. 7

A paper mill produces two grades of wall paper  $W_1$  and  $W_2$ . Because of raw materials restriction,  $W_1$  and  $W_2$  are produced in a limited quantity. At most 400 tons of grade  $W_1$  and at most 300 tons of grade  $W_2$  can be produced in a week. There are 1600 production hours in a week. It requires 2 and 4 hours to produce a ton of product  $W_1$  and  $W_2$  respectively. The profit per ton of  $W_1$  and  $W_2$  are Rs. 2,000 and Rs. 5,000 respectively. Formulate the problem and use simplex method to maximize the profit? [20]

Ans: Max  $Z = 1900000$ ;  $x_1 = 200$ ;  $x_2 = 300$

6. 2067 Q.No. 7 (Old)

Solve the following minimization problem by using simplex method.

Minimize the cost = Rs.  $200x_1$  + Rs.  $800x_2$

Subject to the limitations

$25x_1 + 50x_2 = 750$ ;  $x_1 \leq 100$ ;  $x_2 \geq 70$  Where  $x_1 \geq 0$  and  $x_2 \leq 0$ .

Ans: Unbounded

7. 2066 Q.No. 9

Obtain the optimum solution using simplex technique

Maximize  $Z = 2x_1 + 4x_2 + 3x_3$

Subject to the constraints,

$3x_1 + 4x_2 + 2x_3 \leq 60$

$2x_1 + x_2 + 2x_3 \leq 40$

$x_1 + 3x_2 + 2x_3 \leq 80$

All variables  $\geq 0$ .

20

Ans:  $X_1 = 0$ ;  $X_2 = 20/3$ ;  $X_3 = 50/3$ ;  $S_1 = 80/3$ ; Max  $Z = 230/3$

8. 2065 Q.No. 7

Solve the following problem using simplex method.

Maximize  $Z = 3x_1 + 2x_2 + 5x_3$

Subject to the constraints,  $x_1 + x_2 + x_3 \leq 9$

$2x_1 + 3x_2 + 5x_3 \leq 30$

$2x_1 - x_2 - x_3 \leq 8$  and  $x_1, x_2, x_3 \geq 0$  [20]

Ans: Max.  $Z = 35$  for  $X_1 = 5$ ,  $X_2 = 0$ ,  $X_3 = 4$

9. 2064 Q.No. 7

A firm produces three products that are presented on three different machines. The time required to manufacture one unit of each of the three products and daily capacity of the machines are given in the table below:

| Machine | Product<br>(time/unit in minutes) |       |       | Machine capacity<br>(m/c per day) |
|---------|-----------------------------------|-------|-------|-----------------------------------|
|         | $P_1$                             | $P_2$ | $P_3$ |                                   |
| $M_1$   | 2                                 | 3     | 2     | 440                               |
| $M_2$   | 4                                 | -     | 3     | 470                               |
| $M_3$   | 2                                 | 5     | -     | 430                               |



Determine the daily number of units to be manufactured of each product. The profit per unit for product  $P_1$ ,  $P_2$ ,  $P_3$  is Rs. 4, Rs. 3 and Rs. 6 respectively. It is assumed that all the products are consumed in the market. [20]

$$\text{Ans: Max } Z = 3200/3; x_1 = 0; x_2 = 380/9; x_3 = 470/3$$

**10. 2063 Q.No. 7**

Maximize the profit = Rs.  $600X_1$  + Rs.  $500X_2$

Subject to:  $15X_1 + 20X_2 \leq 600$

$$10X_1 + 5X_2 \leq 200$$

$$X_1 \geq 10$$

$$5X_1 + 5X_2 = 150 \text{ where } X_1, X_2 \geq 0$$

$$\text{Ans: } X_1 = 10, X_2 = 20 \text{ max } (z) = \text{Rs. } 16,000$$

**11. 2060 Q.No. 7**

A furniture company produces two types of products chair and table. The profit contribution on each chair is Rs. 400 and on each table is Rs. 320. The company can sell all the chairs or all the tables or any combination of tables and chairs that it can produce. Unfortunately the production capacities are severely restricted in several respects. First, a special wood, which is used as primary raw material in producing both chairs and tables, is available only in very limited quantities. The labour of highly skilled nature is required in wood working process and also in finishing process. Both types of labour are in scarce supply, so the company can produce only limited quantities of chairs and tables. Each chair requires in its construction 20 units of special wood, while each table require 10 units and maximum quantity of special wood, available is 300 units. Only 100 man hours of wood working is available. Each chair requires 4 hours of wood working labor and 10 hours of wood working labour is required to produce one table. The final step in the production process, the finishing work is also performed by highly skilled labour, and a maximum of 38 man-hours of this finishing labour is available. Each chair required two man-hours and each table three man-hours of finishing time. Formulate a linear programming model and determine how much of each product should be manufactured to maximize total profit contribution by using simplex method. [20]

$$\text{Ans: } X_1 = 13 \text{ units; } X_2 = 4 \text{ units; max } (z) = \text{Rs. } 6,480$$

**12. 2059 Q.No. 7**

Define 'degeneracy' in linear programming model and find the optimal solution for the following problem by using simplex. [20]

Maximize the profit = Rs.  $200A$  + Rs.  $600B$  + Rs.  $80C$

Subject to

$$16A + 4B + 6C \leq 500$$

$$4A + 3B + 0C \leq 150$$

$$2A + 0B + C \leq 50$$

$$A, B, C \geq 0$$

$$\text{Ans: } A = 0; B = 50; C = 50, \text{ max. profit} = \text{Rs. } 34,000$$

**MBA****13. 2056 Q.No. 4**

A furniture company produces table, chair and book-cases and all of the produces have to go through assembly, finishing and packing departments. The management has formulated the maximization problem as follows:

$$\text{Max. } Z = 2T + 4C + 3B$$

Subject to:  $3T + 4C + 2B \leq 60$  (assembly constraints)

$$2T + 4C + 2B \leq 40 \text{ (Finished constraints)}$$

$$1T + 3C + 2B \leq 80 \text{ (packing constraints)}$$

All variable  $\geq 0$

Solve above linear programming problem.

$$\text{Ans: Table} = 0; \text{Chair} = 20/3; \text{Bookcase} = \text{Rs. } 50/3; \text{Max } (z) = \text{Rs. } 230/3$$

**14. 2055 Q.No. 6**

Use the simplex method to find the value of  $x_1$  and  $x_2$  which maximizes the function:

$$P = -2x_1 - 8x_2 \text{ for the following:}$$

$$5x_1 + 10x_2 = 150$$

$$0 \leq x_1 \leq 20$$

$$x_2 \geq 14$$

$$\text{Ans: } X_1 = 2 \text{ units, } X_2 = 14 \text{ units, max } (z) = \text{Rs. } -116$$



**15. 2054 Q.No. 6**

An investor what to allocate his portfolio among five different types of securities. The expected returns, denoted by  $E(R)$  and standard deviations of expected return, denoted by  $\sigma$  (s.d.) of five different types of securities are as given below:

| Risk return characteristics |            |        |          |
|-----------------------------|------------|--------|----------|
| Types of securities         | Proportion | $E(R)$ | $\sigma$ |
| Speculative stock           | $X_1$      | 14%    | 10%      |
| Mutual fund                 | $X_2$      | 10%    | 5%       |
| Preference stock            | $X_3$      | 8%     | 4%       |
| Long term bond              | $X_4$      | 7%     | 2%       |
| Bank savings account        | $X_5$      | 5%     | 0%       |

The investor's objective is to maximize his expected return, although the realization of this objective must be conditioned by the existence of several constraints which are as follows:

- He cannot allocate more than 100% of his total portfolio.
- He is also unwilling to hold a portfolio with a combined standard derivation in excess of 6%.
- The proportion invested in speculative stock must be limited to 0.4.
- He cannot invest negative amounts in any types of security.

Solve the problem by simplex method.

[20]

Ans:  $X_1 = 0.2$ ;  $X_2 = 1.2$ ;  $X_3 = 0$ ;  $X_4 = 0$ ;  $X_5 = 98$ , Maximum expected return = 506.

**16. 2052 Q.No. 5**

A publisher sells a deluxe hardcover and paperback edition of the same textbook for Rs. 15, Rs. 9 and Rs. 5 respectively. Costs to the publisher are Rs. 12, Rs. 7 and Rs. 3.60 per book respectively.

Deluxe requires 8 minutes of printing time while other two require 5 minutes each only. Deluxe cover requires 12 minutes of binding time but hardcover and paperback requires 10 and 2 minutes respectively. Both the printing and binding operations have 4800 minutes available each week. How many of each types of books should be produced in order to maximize profit? [20]

Ans:  $X_1 = 3600/11$  units;  $X_2 = 0$ ;  $X_3 = 4800/11$  units, Max (z) = Rs. 17520/11

**17. 2051 Q.No. 5**

A company makes three products, A, B and C, each of which is produced from three main elements,  $E_1$ ,  $E_2$  and  $E_3$ . The products are packed in 100 kgs. The contribution for a bag of A is Rs. 30, for a bag of B Rs. 50 and or a bag C Rs. 40. Each product utilizes the number of pound of each elements shown.

| Product | Element |       |       |
|---------|---------|-------|-------|
|         | $E_1$   | $E_2$ | $E_3$ |
| A       | 2       | 0     | 3     |
| B       | 3       | 2     | 2     |
| C       | 0       | 5     | 4     |

There are following limits on the amounts of the elements available.

$E_1$  - 800 kgs.;  $E_2$  - 1,000 kgs.;  $E_3$  - 1,500 kgs.

How will you determine how much of each product should be produced to maximize contribution? [20]

Ans: A = 8900/41; B = 5000/41; C = 6200/41, max (Z) = Rs. 765,000/41

**18. 2050 Q.No. 5**

Bentwood Manufacturing Company produces a bent wood rocking chair and a bentwood coffee table, each of superior styling and quality. The profit contribution on each chair is Rs. 40 and on each table is Rs. 32. The company can sell at the prevailing prices either all the chairs or all the tables, or any combination of chairs and tables that it can produce.

Unfortunately, the production capabilities of the Bentwood plant are severely restricted in several respects. First, a special rosewood which is used as the primary raw material in both the chairs and the tables is available only in very limited quantities. Then, labour of highly skilled nature is required in the wood working process and also in the finishing process. Both types of labour are in scare supply. Burdened by these constraints, the company can



manufacture only limited quantities of chairs and tables. Each chair requires in its construction 40 units of rosewood, while each table requires 20 units of rosewood. The only known supplier of rosewood can furnish Bentwood with a maximum of 600 units of wood each day. Only 100 man-hours of wood working are available to Bentwood each day. Each chair requires 4 hours of wood working labour, and each table requires 10 hours of this labour.

The final step in the production process, the finishing work, is also performed by highly skilled employees, and a maximum of 38 man-hours of this finishing labour is available each day. Each chair requires two man-hours and each table, three man-hours of finishing time. Formulate a linear programming model and determine how much of each product should be manufactured to maximize total profit contribution by using simplex method. [20]

Ans: Chair ( $X_1$ ) = 13 units Table ( $X_2$ ) = 4 units, Max ( $z$ ) = Rs. 648

#### 19. 2048 Q.No. 5

The XYZ Company combines factors A and B to form a product, which must weight 50 pounds. At least 20 pounds of A and no more than 40 pounds of B can be used. A costs Rs 25 per pound, and B costs Rs. 10 per pound.

Use the simplex method to find the amounts of factor A and factor B which should be used to minimize the cost.

Ans: A = 20 units; B = 30 units; Cost ( $c$ ) = Rs. 800

#### 20. 2046 Q.No. 5

A manufacturing company manufactures two different products. The demand for both the products is strong enough so that the firm can sell as many units of either product, or of both; as it can produce and at such a price as to realize a per unit profit contribution of Rs. 16 on product A and Rs. 10 on product B. Unfortunately, the production capacity of the company's plant is severely limited. This limitation stems from the fact that the manufacture of the products involves the utilization of three scarce resources: raw material, labour and machine time. Each unit of product A requires four units of raw material, three units of labour and two units of machine time. Each unit of product B requires two units of raw material, three units of labour and five units of machine time. The firm has a daily supply of 24 units of raw material, 21 units of labour and 30 units of machine time.

Formulate a linear programming model and determine how much of each product should be manufactured to maximize total profit contribution by using simplex method. [20]

Ans: A = 5 units; B = 2 units; C = 0 units; Max ( $z$ ) = Rs. 100

#### 21. 2045 Q.No. 5

Solve the following problem through linear programming problem

Maximize  $Z = \text{Rs. } 2A + \text{Rs. } 4B + \text{Rs. } 3C$

Subject to

$$3A + 4B + 2C \leq 60$$

$$2A + B + 2C \leq 40$$

$$A + 3B + 2C \leq 80$$

$$A, B, C \geq 0$$

[20]

Ans: A = 0; B = 20/3; C = 50/3, Max ( $z$ ) = Rs. 230/3

#### 22. 2042 Q.No. 5

Solve the following problems by simplex method.

Maximize:  $Z = x_1 - x_2 + 3x_3$

Subject to  $x_1 + x_2 + x_3 \leq 10$

$$2x_1 - x_3 \leq 2$$

$$2x_1 - 2x_2 + 3x_3 \leq 6$$

$$x_1, x_2, x_3 \geq 0$$

[20]

Ans:  $X_1 = 0$ ,  $X_2 = 24/5$ ,  $X_3 = 26/5$ ,  $S_2 = 36/5$ , Max ( $z$ ) = Rs. 54/5

#### 23. 2041 Q.No. 5

The XYZ Company combines factors A and B to form a final product which must weight exactly 150 pounds. Factors A cost Rs. 2 per unit and B cost Rs. 8 per unit. At least 14 units of B and no more than 20 units of A must be used. Each unit of A weights 5 pounds and each unit of B weight 10 pounds. How much each type of raw materials should be used for each unit of B weight 10 pounds. How much each type of raw material should be used for each unit of final product if company wish to minimize cost?

[20]

Ans: Cost ( $C$ ) = Rs. 116 when  $X_1 = 2$  units and  $X_2 = 14$  units



**24. 2040 Q.No. 8**

Solve the following linear programming problem

Maximize  $Z = \text{Rs. } 4A + \text{Rs. } 8B + \text{Rs. } 6C$

Subject to

$$6A + 8B + 4C \leq 120$$

$$4A + 2B + 4C \leq 80$$

$$A + 6B + 4C \leq 160 \quad A, B, C \geq 0 \quad [20]$$

Ans:  $A = 0$ ;  $b = 20/3$ ;  $C = 50/3$ ;  $\text{Max}(z) = \text{Rs. } 460/3$

**25. 2039 Q. No. 8**

A company makes three products, A, B and C, each of which is produced from three main elements,  $E_1$ ,  $E_2$  and  $E_3$ . The products are packed in 100 kgs. The contribution for a bag of A is Rs. 30, for a bag of B Rs. 50 and or a bag C Rs. 40. Each product utilizes the number of pound of each elements shown.

| Product | Element |       |       |
|---------|---------|-------|-------|
|         | $E_1$   | $E_2$ | $E_3$ |
| A       | 2       | 0     | 3     |
| B       | 3       | 2     | 2     |
| C       | 0       | 5     | 4     |

There are following limits on the amounts of the elements available.

$E_1$  - 800 kgs.;

$E_2$  - 1,000 kgs.;

$E_3$  - 1,500 kgs.

How will you determine how much of each product should be produced to maximize contribution? [20]

Ans:  $X_1 = 8900/41$ ;  $X_2 = 5000/41$ ;  $X_3 = 6200/41$ ; maximum contribution (z) = Rs. 765,000/41